

Chemical Reference Document

A comprehensive collection of chemical compounds

Aromatic Hydrocarbons:

Benzene (C₆H₆) - c1ccccc1

Toluene (C₇H₈) - Cc1ccccc1

Phenol (C₆H₆O) - Oc1ccccc1

Aniline (C₆H₇N) - Nc1ccccc1

Naphthalene (C₁₀H₈) - c1ccc2ccccc2c1

Nitrobenzene - O=[N+]([O-])c1ccccc1

Benzoic acid - OC(=O)c1ccccc1

Pharmaceuticals:

Aspirin (C₉H₈O₄) - CC(=O)Oc1ccccc1C(=O)O

Caffeine (C₈H₁₀N₄O₂) - Cn1c(=O)c2c(ncn2C)n(C)c1=O

Ibuprofen - CC(C)Cc1ccc(cc1)C(C)C(=O)O

Paracetamol (Acetaminophen) - CC(=O)Nc1ccc(O)cc1

Morphine - CN1CCC23c4c5ccc(O)c4OC2C(=CC1C3C5)O

Penicillin G - beta-lactam antibiotic

Nicotine - CN1CCCC1c1cccnc1

Heterocyclic Compounds:

Pyridine - c1ccncc1

Quinoline - c1ccc2ncccc2c1

Furan - c1ccoc1

Thiophene - c1ccsc1

Pyrrole - c1cc[nH]c1

Indole - c1ccc2[nH]ccc2c1

Nucleobases:

Adenine - Nc1ncnc2[nH]cnc12

Guanine - nucleobase of DNA

Cytosine - Nc1ccn(c(=O)n1)

Thymine - Cc1cn(c(=O)[nH]c1=O)

Uracil - O=c1cc[nH]c(=O)[nH]1

Steroids and Hormones:

Cholesterol - steroid compound

Testosterone - steroid hormone C19H28O2

Estradiol - estrogen hormone

Cortisol - glucocorticoid hormone

Progesterone - progesterone hormone

Neurotransmitters:

Dopamine - NCCc1ccc(O)c(O)c1

Serotonin - NCCc1c[nH]c2ccc(O)cc12

Adrenaline (Epinephrine) - fight-or-flight hormone

Sugars and Carbohydrates:

Glucose - OCC1OC(O)C(O)C(O)C1O

Fructose - fruit sugar

Sucrose - table sugar disaccharide

Terpenes and Natural Products:

Vanillin - O=Cc1ccc(O)c(OC)c1

Cinnamaldehyde - O=C/C=C/c1ccccc1

Limonene - CC(=C)C1CCC(=CC1)C

Menthol - CC1CCC(C(C1)O)C(C)C

Camphor - monoterpene ketone

Retinol - Vitamin A

Ascorbic acid - Vitamin C

Simple Organic Compounds:

Acetone - CC(=O)C

Ethanol - CCO

Acetic acid - CC(=O)O

Methanol - CO

Glycine - NCC(=O)O

Alanine - CC(N)C(=O)O